EXP3

April 10, 2026

[6]:
```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_openml
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
```

[10]:
```
mnist = fetch_openml('mnist_784', version=1, as_frame=False)

X = mnist.data
y = mnist.target.astype(int)

print(X.shape)
```

```
(70000, 784)
```

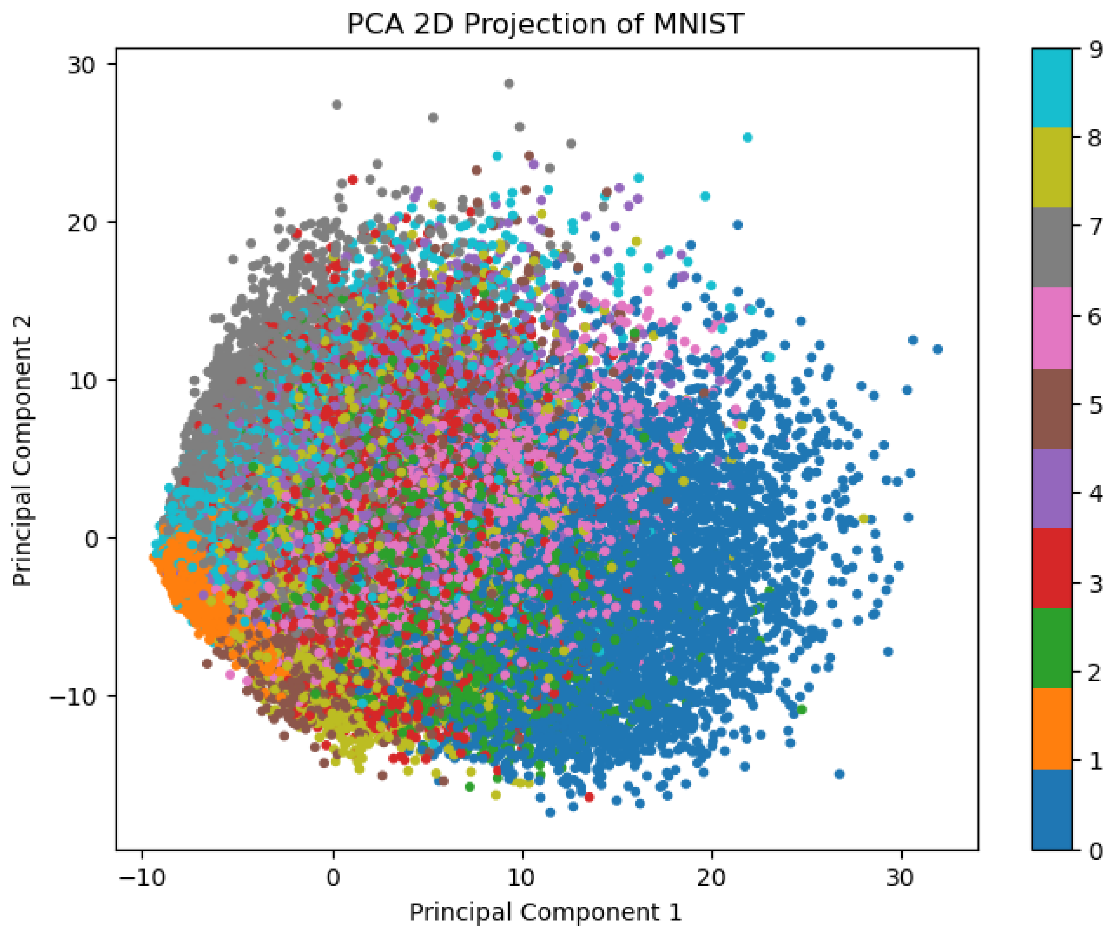
[12]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

[16]:
```
pca_2d = PCA(n_components=2)
X_pca_2d = pca_2d.fit_transform(X_scaled)
```

[18]:
```
plt.figure(figsize=(8,6))

scatter = plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    c=y,
    cmap='tab10',
    s=10
)

plt.colorbar(scatter)
plt.title("PCA 2D Projection of MNIST")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()
```



[20]:

```
pca_50 = PCA(n_components=50)
X_pca_50 = pca_50.fit_transform(X_scaled)

print("Reduced shape:", X_pca_50.shape)
```

```
Reduced shape: (70000, 50)
```

[22]:

```
pca_full = PCA()
pca_full.fit(X_scaled)

explained_variance = pca_full.explained_variance_ratio_
cumulative_variance = np.cumsum(explained_variance)
```

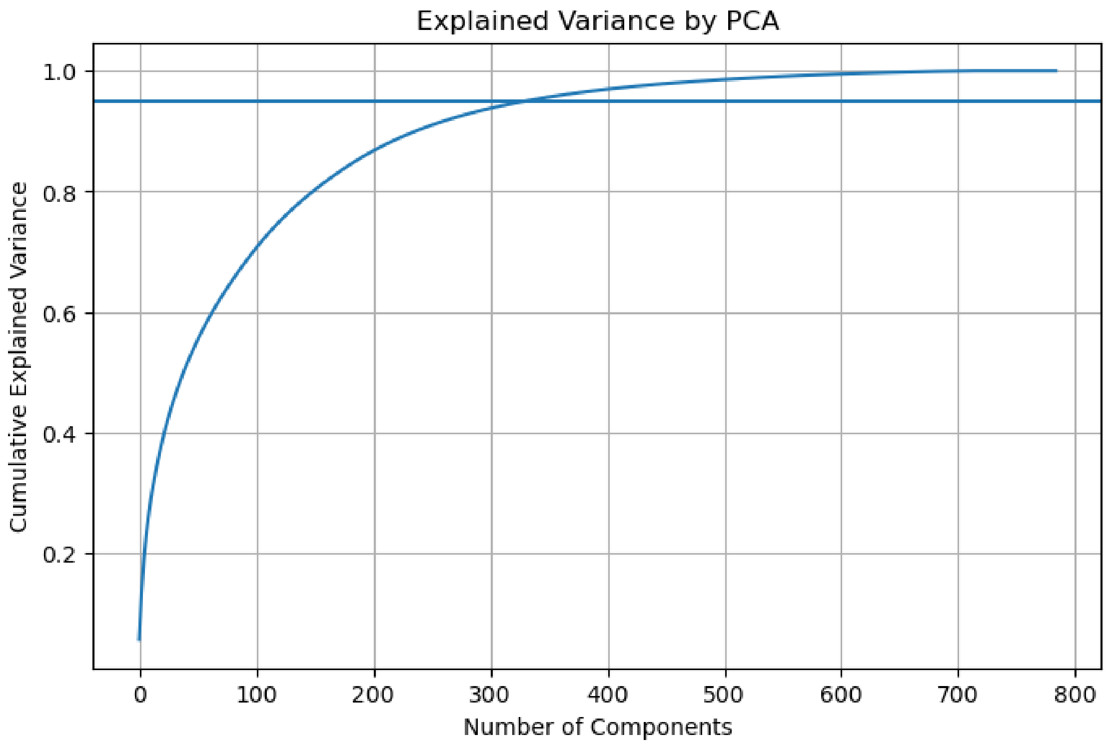
[28]:

```
plt.figure(figsize=(8,5))

plt.plot(cumulative_variance)
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
```

```python
plt.title("Explained Variance by PCA")
plt.grid()

plt.axhline(y=0.95)
plt.show()
```



[26]:
```python
n_components_95 = np.argmax(cumulative_variance >= 0.95) + 1
print("Components needed for 95% variance:", n_components_95)
```

```
Components needed for 95% variance: 332
```

[ ]:
```python

```